CS 4412 M4 Data Mining Project

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I. Dataset Description

My main datasets are the IMDb top 250 movies and Box Office Mojo's top 200 highest grossing movies, with Rotten Tomatoes acting as a bridge between them. Each of my sources represents the main points of my discovery questions; IMDb has the general response, BOM has the profits, and RT has the individual audience and critic responses. An important thing to note about the datasets is that not every movie is shared between them, as in some films were hated but profitable while others became forgotten classics. Instead of fixing this gap, I plan to embrace it. After all, blockbusters and critical darlings are the lifeblood of the industry.

The end result is a catalog of around 450 movies of various responses and profits. Each movie will have 5 columns. The main three are their box office, their critical reception, and their audience receptions. These categories and the ways they intersect are what I am analyzing for my project. I have included titles and year of release just for the sake of formatting.

Frozen II	$1,453,683,476	77%	92%	2019
Léon: The I	$20,330,788	76%	95%	1994
The Chronicles of Narnia: The Lion, the Witch and the Wardrobe	$745,013,115	76%	61%	2005
Harry Potter and the Deathly Hallows: Part 1	$960,858,478	76%	85%	2010
Avatar: The Way of Water	$2,334,484,620	76%	92%	2022
Lock, Stoc	$3,753,929	75%	93%	1998
The Intouc	$426,590,315	75%	93%	2011
Forrest Gump	$678,226,465	75%	95%	1994
Forrest Gump	$678,226,465	75%	95%	1994

Fig. 1. Raw Dataset. Note the duplicate at the bottom.

II. Discovery Questions

The big questions driving my project are:
1. "What kind of films emerge based on the critic and audience opinion?"
2. "What mixture of ratings and revenue distinguishes a 'cult classic' from a 'blockbuster'?"
3. "How do good reviews keep a movie alive in the public consensus?"

I wanted to ask these questions because of my extended contact with the media discussion side of the internet, for better or worse. I have seen people claim that a highly profitable film with good critical reviews had bribed reviewers because it had low audience ratings, and I have seen those same people claim that a profitable film with low critical ratings has a cult audience blindly giving the media their money. This resulted in the first two questions being asked to find a solid answer. Besides that, I also wanted to discuss There is a term known as "Vindicated by History", where a work that failed in the former present becomes heavily praised in the current present. I was reminded of this phrase after I looked through my datasets and found that a sizable number of famous movies do not make that much of a profit. So, I decided to establish another study on the differences between financial impact and cultural impact.

With the implementation finished, I can finally answer these questions based on what I have found.

A. *Question 1: Movie Patterns*

The four main patterns of movies are Excellent, Biased, Mixed, and Panned. Excellent Movies are given high praise from both group, regardless of how many members went out to see them in theaters. Biased Movies are given good treatment from both groups but one side enjoys it more. Mixed Movies have good ratings from one side and middling reviews from the other. Panned Movies are negatively received from both types of film-goers.

B. *Question 2: Profit Patterns*

The line between a smash hit and a stealth success is small in length but large in width. The length of the line takes the form of positive reception, which can be gotten from every critic enjoying an unpopular film or from most critics liking a famous film. Meanwhile, the width of the divider takes the form of negative reception. Since opinions only form after watching the movies, bad opinions feed the box office. Cult classics that are enjoyed by the few who see them and blockbusters that attract millions and print millions more are separated by movies that are liked by enough groups of people to make a profit.

C. *Question 3: Longevity*

Good reviews can keep older movies alive in the mind of the public much more than the box office can. This can manifest in multiple ways. Sometimes it comes in the form of franchises or inspired media, but that has the major risk of producing something that audiences don't enjoy. Other times, it is through releases. This is why there is a single spike in profitable movies around 1939. The most important way reviews fuel old movies is the way invoked in the previous question- cult classics that people enjoy watching and talk about it to others so that they can watch it to.

III. Exploratory Data Analysis

Originally, I wanted to create a punnet square to sort the movies according to their audience and reviewer scores. This would have reflected the Rotten Tomatoes scoring system but not simulate it, as I planned for it to follow my k-means results However, after going through EDA, I have chosen to follow some more solid graphs.
Once all of the films have been recorded and defined, the real analysis begins. I have a few plots for each column that help establish some findings. The box-plots show that most movies earn under a billion dollars, usually half that, with the major money makers being extreme outliers. Meanwhile, critics and audiences tend to enjoy what they watch but have different standards for what they tend to enjoy.

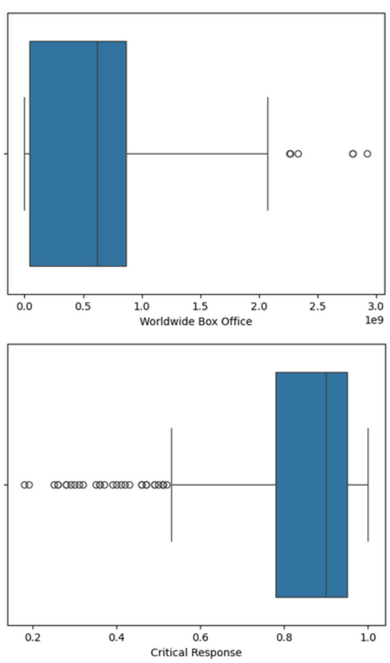

Fig. 2. Profit and Critical Response Box Plots

Meanwhile, the bar graphs show just how big the gap between profit and popularity really is. The number of low-profit movies compared to everything else, even the ones in the hundreds of millions, combined with the peak reviews being in the good to great range implies the preference of classics over blockbusters. However, my data could have been heavily twisted by outside factors such as the demographic of moviegoers across the years or people's personal biases affecting the total reception.

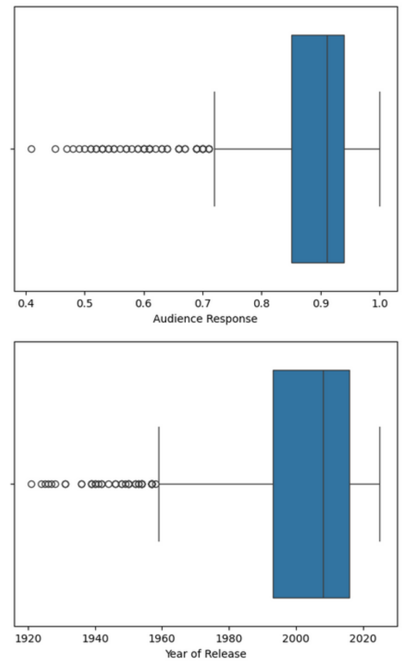

Fig. 3. General Response and Release Year Box Plots

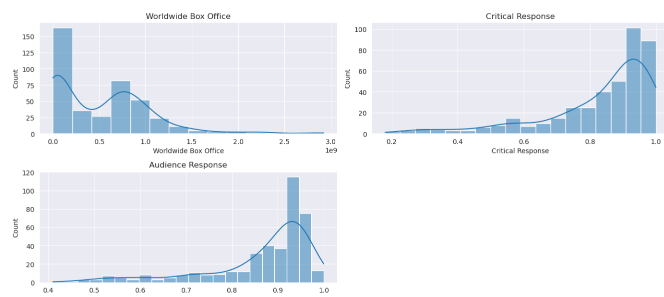

Fig. 4. Average Distribution of Box Office (Top Left), Critical Response (Top Right), and Audience Response (Bottom)

A. Data Preprocessing

Due to taking data from two separate sites that discuss the profits and reception of movies respectively, there were a few duplicates. An interesting thing to note is that most of the duplicates had some kind of brand attachment, like Pixar for *Finding Nemo* or Tom Hanks for *Forrest Gump*. The *Fast and Furious* franchise lacking any repeats while the *Jurassic Park* series only has the first film duplicated implies a required amount of skill and marketing to appear on both lists.
As for true outliers, the only major one is a 1975 Turkish comedy named *The Chaos Class Failed the Class*, which appears to be a complete anomaly. The minor outliers has a missing box office number or critical reception, but these can be fixed through third party sources.

B. Overlap

Looking at where the data point clump together forms a really intriguing picture. When comparing profits and critical reception, there appears to be a universal standard profit or 800 million dollar that is formed across all metrics of reviews.

```
#print(df.head())
#df.info()
#df.nunique()
#df.isnull().sum()
#df = df.dropna()
#df.isnull().sum()
#df = df.drop_duplicates()
df.describe()
```

	Worldwide Box Office	Critical Response	Audience Response	Year of Release
count	444.00	443.00	446.00	447.00
mean	554,004,154.43	0.84	0.87	1,999.77
std	520,125,937.65	0.17	0.12	23.46
min	399.00	0.18	0.41	1,921.00
25%	39,534,711.75	0.78	0.85	1,993.00
50%	620,358,997.50	0.90	0.91	2,008.00
75%	864,076,097.00	0.95	0.94	2,016.00
max	2,923,710,708.00	1.00	1.00	2,025.00

Fig. 5. Data before Preprocessing

```
#print(df.head())
#df.info()
#df.nunique()
#df.isnull().sum()
df = df.dropna()
#df.isnull().sum()
df = df.drop_duplicates()
df.describe()
```

	Worldwide Box Office	Critical Response	Audience Response	Year of Release
count	411.00	411.00	411.00	411.00
mean	519,749,158.78	0.83	0.87	1,998.90
std	507,047,335.53	0.17	0.12	23.99
min	399.00	0.18	0.41	1,921.00
25%	33,007,748.00	0.77	0.84	1,989.50
50%	482,352,390.00	0.90	0.91	2,008.00
75%	830,237,770.50	0.94	0.94	2,016.00
max	2,923,710,708.00	1.00	1.00	2,025.00

Fig. 6. Data after Preprocessing

In the box graphs for year of release, films has slowly risen their profit limits so that what was the maximum in the 80s or 90s is the standard of the 2010s. Meanwhile, the increase in film releases means that more underwhelming films are being watched as well. Finally, and once again, audiences and critics can agree on what is excellent but disagree on what is mediocre.

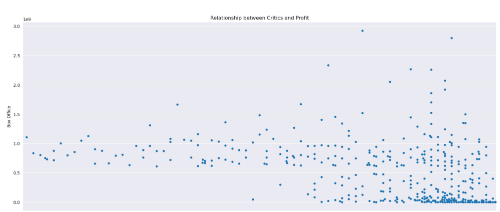

Fig. 7. Swarm Plot of Critics versus Profits

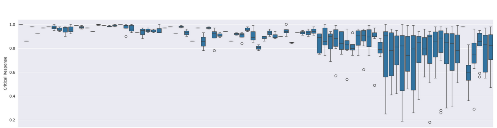

Fig. 8. Box Graph of Year of Release versus Critical Response

IV. Methods

The two technique used to find the data were K-means clustering and decision trees. Clustering was useful in establishing a foundation and detecting what patterns have emerged from the data. Decision trees supported the centroids by working backwards from the clusters to find where the data points began to fracture. The reason I chose this approach was due to its ease of use, which complimented my data and personality well.

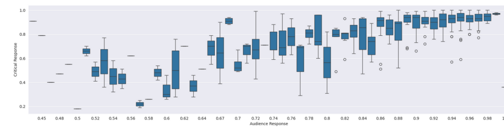

Fig. 9. Box Graph of Year of General versus Professional Responses

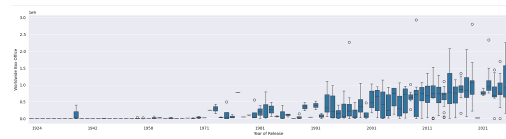

Fig. 10. Box Graph of Year of Release versus Box Office

I started by using clustering to mine through my datasets to help smooth down the rough data into my precise rankings. I built a k-means algorithm to find the optimal number of clusters. My elbow-graph result found three to be the best match, so I plugged that amount of clusters into the model and ran it ten times. Two of the clusters were overlapping each other, which I ignored until I tried to find the centroids. After cross-referencing my two kmeans functions with my elbow graph, I realized that there might have been *4* clusters in my data. After looking over the results and my analysis, I have

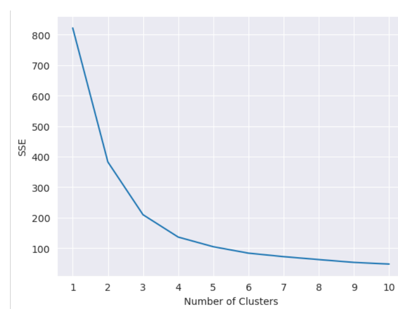

Fig. 11. K-Means Elbow Graph

settled on the data having 4 clusters. Part of the reason why was that the standardized data being cleanly split between the 4 clusters while having k=3 created a stretched-out blob of a cluster. Another reason was how the resulting patterns formed a much clearer picture in k=4 than in k=3.

For M3, I expanded into decision trees to further strengthen my data. The idea was to split the data by the box office and see what patterns of reception can be formed. For example, I could split the data by less than 1 billion dollars in profit and calculate the gain achieved from that. However, the data couldn't function right and I ended up having to shift to reverse engineering the formation of the clusters.

The first thing I noticed was how it immediately went to compare the box office of the data, and at such a low value.

The films that missed the profit goal were then judged by their release year, since most of the low-performing data were part of the Classic or Genre clusters. Films that did exceed the profit goal were compared again to a much higher attribute, which split the results into Blockbuster movies and Minivan movies. Back to the low profit branch, the data was split between before and after 1977. The Classic cluster was finally established at this point while the remaining data had to be compared against worsening box office.

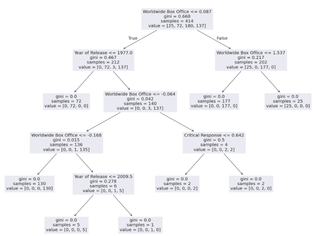

Fig. 12. Decision Tree

V. Results

A. Clusters

The first cluster, dubbed 'Blockbuster', is similar to the Minivan cluster in how the data is defined by profit first and responses second. What separates the two film clusters is that people don't just enjoy blockbusters, they appreciate them. Blockbusters hold the highest combination of audience, critical, and box office reception across the entire dataset. This is caused by a multitude of factors across all metrics. The data of the cluster is spearheaded by the outliers with over 2 billion dollars in profit. Meanwhile, a notable chunk of the data hold major sway over the viewer due to a relation with a wide-reaching franchise like Marvel or a director like James Cameron. Related to this, most of the data points were released during the 21st century when budgets became overloaded and the industry went all in. There are around 25 blockbuster movies according to the decision tree.

The second cluster, dubbed 'Classic', holds movies that everyone remembers due to pop culture osmosis. They are marked by extremely high responses from critics, with audiences being secondary priority despite their support. Another important sign of a classic is the severe dip in box office profits. The age of the data might be why the box office is so low since inflation hasn't been applied to those data points yet. I have chosen not to calculate the box office in the context of inflation due to the precedence it set for all data points as well as the excess work involved. There are 72 classic movies based on both the decision trees and the k-means summary.

The third cluster, dubbed 'Minivan', consists of movies that are usually seen by groups of people. These films are meant for multiple different audiences, such as families with children or teens who watch it as a joke. Minivan movies tend to prefer profits over reception, reaching a consistent box office of 1 billion dollars regardless of how critics or even the general

	Film Name	Worldwide Box Office
6	Avatar: Fire and Ash	1.91
20	Ne Zha 2	3.46
21	Spider-Man: No Way Home	2.77
26	Zootopia 2	2.66
27	Deadpool & Wolverine	1.62
34	Inside Out 2	2.33
45	Barbie	1.83
60	The Super Mario Bros. Movie	1.66
65	Avatar: The Way of Water	3.59
79	Top Gun: Maverick	1.93
101	Avengers: Endgame	4.51
109	Frozen II	1.85
125	The Lion King (2019)	2.26
134	Avengers: Infinity War	3.03
137	Black Panther	1.64
149	Jurassic World: Fallen Kingdom	1.56
178	Star Wars: Episode VIII - The Last Jedi	1.61
220	Avengers: Age of Ultron	1.75
223	Furious 7	1.97
228	Jurassic World	2.28
239	Star Wars: Episode VII - The Force Awakens	3.07
310	The Avengers	1.98
325	Harry Potter and the Deathly Hallows: Part 2	1.63
362	Avatar	4.76
508	Titanic	3.45

	Critical Response	Audience Response	Year of Release	cluster
6	-0.99	0.29	2025	0
20	0.47	0.97	2025	0
21	0.58	0.89	2025	0
26	0.47	0.80	2025	0
27	-0.29	0.63	2024	0
34	0.47	0.63	2024	0
45	0.29	-0.30	2023	0
60	-1.39	0.72	2023	0
65	-0.40	0.46	2022	0
79	0.76	1.06	2022	0
101	0.64	0.29	2019	0
109	-0.35	0.46	2019	0
125	-1.80	0.12	2019	0
134	0.12	0.46	2018	0
137	0.76	-0.64	2018	0
149	-2.09	-3.27	2018	0
178	0.47	-3.87	2017	0
220	-0.46	-0.39	2015	0
223	-0.11	-0.39	2015	0
228	-0.64	-0.73	2015	0
239	0.58	-0.22	2015	0
310	0.47	0.38	2012	0
325	0.76	0.21	2011	0
362	-0.11	-0.39	2009	0
508	0.29	-1.49	1997	0

Fig. 13. Blockbuster data subset

	Film Name	Worldwide Box Office	Critical Response
575	Network	-0.99	0.47
576	Rocky	-0.80	0.58
577	Taxi Driver	-0.97	0.35
578	Barry Lyndon	-1.03	-0.29
579	Jaws	-0.06	0.82
..	...	...	...
643	Metropolis	-1.03	0.82
644	The General	-1.03	0.53
645	The Gold Rush	-1.03	0.87
646	Sherlock Jr.	-1.03	0.18
647	The Kid	-1.03	0.99

	Audience Response	Year of Release	cluster
575	0.55	1976	1
576	-1.49	1976	1
577	0.55	1976	1
578	0.46	1975	1
579	0.38	1975	1
..	...	...	...
643	0.46	1927	1
644	0.46	1926	1
645	0.55	1925	1
646	0.72	1924	1
647	0.72	1921	1

[72 rows x 6 columns]

Fig. 14. Classic data subset

public thought of them. This has the consequence of Minivan movies being seen as forgettable at best or infamous at worst due to moviegoers not getting engaged by the results. There are at least 177 minivan movies with a few scattered outliers between the various records.

	Film Name	Worldwide Box Office
1	Project Hail Mary	0.23
2	The Super Mario Galaxy Movie	0.45
4	A Minecraft Movie	0.87
9	Demon Slayer: Kimetsu no Yaiba Infinity Castle	0.42
10	Demon Slayer: Kimetsu No Yaiba Infinity Castle	0.44
..	...	...
522	Forrest Gump	0.31
527	The Lion King	0.91
532	Jurassic Park	1.15
560	E.T. the Extra-Terrestrial	0.55
572	Star Wars: Episode IV - A New Hope	0.50

	Critical Response	Audience Response	Year of Release	cluster
1	0.64	0.72	2026	2
2	-2.32	0.21	2026	2
4	-2.09	-0.22	2025	2
9	0.87	0.97	2025	2
10	0.87	0.97	2025	2
..	...	...	...	...
522	-0.46	0.72	1994	2
527	0.53	0.55	1994	2
532	0.47	0.38	1993	2
560	0.93	-1.24	1982	2
572	0.64	0.80	1977	2

[180 rows x 6 columns]

Fig. 15. Minivan data subset

The finale cluster, dubbed 'Genre', is populated by well-received movies with very specialized aesthetics. They have the critical reception of the classic cluster and the profits of the minivan cluster, just not to the same extent or consistency. The audience scores varies across films, but the nature of genre movies makes this a minor concern at best. The more profitable side of the cluster something to attract viewers towards them, such as a famous brand or a well known director. There are 135 films, give or take some critically panned outliers, as shown on the decision tree.

```
                            Film Name  Worldwide Box Office  \
7    Chainsaw Man - The Movie: Reze Arc                 -0.71
36                             Maharaja                 -1.03
41                       The Wild Robot                 -0.37
44                            12th Fail                 -1.03
96                             Hamilton                 -0.99
..                                  ...                   ...
567                    The Elephant Man                 -0.98
568                          The Shining                 -0.93
569                                Alien                 -0.82
570                       Apocalypse Now                 -0.82
571                      The Deer Hunter                 -0.93

      Critical Response   Audience Response  Year of Release  cluster
7                  0.76                0.97             2025        3
36                 0.00                0.63             2024        3
41                 0.82                0.97             2024        3
44                 0.47                0.89             2023        3
96                 0.87                1.06             2020        3
..                  ...                 ...              ...      ...
567                0.47                0.55             1980        3
568                0.06                0.55             1980        3
569                0.58                0.63             1979        3
570                0.47                0.63             1979        3
571                0.18                0.38             1978        3

[137 rows x 6 columns]
```

Fig. 16. Genre data subset

B. Centroid

The main standardized data shows the 4 clusters very well. The black curve in the top right represents the blockbuster cluster with its high profits and high reception. The orange blob in the middle is the minivan cluster, which only has moderate box office and a wider range of review scores. The maroon colored line along the bottom left are the classic films, which have been skewed by low box office scores and differences between critics and audiences. The beige data points between the classics and minivan clusters is the genre cluster, which mixes the reception of the latter with the profits of the former.

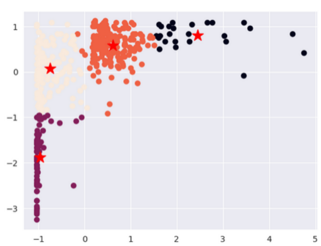

Fig. 17. K-Means Centroid Graph

VI. Assessment

I feel very confident about what I have worked on and written about. The results and analysis make sense both within the confines of my data and in the context of how the movie industry as a whole works. The data itself was formed by cross-referencing multiple verified databases that are specialized in cataloging info on movies and similar properties. Most importantly, I have been experimenting with this data for months without any issues or corruptions. I have tested multiple different methods and have found the systems that work the best for me. All in all, I would say that this report and the project as a whole are quite valid in its findings.

The biggest limitation of this project is its scope and scale. The compressed structure of the data set means that there is a limited amount of ways that one could process the information without adding more feature sets to it, but that brings in the problem of scope creep and comparisons that don't add anything to the wider picture. As for the data by itself, it all comes from multiple sources that are updated constantly with more reviews, updated statistics, and even new releases. As such, the dataset must be either be outdated and only set to a specific date, or regularly updated with the sources' relevant information.

There is nothing ethical concerning about reviewing movies, unless you count indirect means like movies starring problematic people or movies about concerning themes. Even then, this project is meant to analyze the reception of movies, regardless of their quality or impact.

VII. Conclusion

To summarize, I will first look back at my discovery questions. Based on the solely the opinion of casual and competitive moviegoers, it is difficult to see what kind of movies start forming. A notable number of films prioritize one audience over another, such as Classic preferring critics or Minivan preferring audiences, but their profits have a slight overlap. The general ideas of the four cluster are shaped by the analytical reception to the movie, but are finalized by it's economical reception. This leads to my next point on what differentiates a cult classic from a blockbuster. The short answer is that either category can have great critical reception, but cult classics are marked by a drop in general reception and profit while blockbusters maintain the same positive results in all three categories. Finally, there is how a movie can survive on good reviews. I can confidently say that this one is self-evident by the mere existence of the Classic category. While some films can get by on box office and average reviews, the higher ranked films are the one celebrated the most.

As for my project as a whole, I feel very confident about what I have worked on and written about. The results and analysis make sense both within the confines of my data and in the context of how the movie industry as a whole works.

I may not return to this idea in the future. I may not work on something like this ever again. But it did help me understand what it means to work on something that you are proud of and that you believe in.

VIII. Sources

Github Link

Griffin Vines, "GitHub - PcktCalc-Kraftwrk/CS4412-Project," GitHub, 2025. https://github.com/PcktCalc-Kraftwrk/CS4412-Project (accessed April. 02, 2026).

Dataset links

IMDb, "IMDb Top Rated Movies," IMDb, 2019. https://www.imdb.com/chart/top/

[2] "Top Lifetime Grosses - Box Office Mojo," Box Office Mojo, 2019. https://www.boxofficemojo.com/chart/ww_top_lifetime_gross/?area=XWW

[3] Rotten Tomatoes, “Rotten Tomatoes: Movies — TV Shows — Movie Trailers — Reviews,” Rotten Tomatoes, 2025. https://www.rottentomatoes.com/

[4] GeeksforGeeks, “Data Preprocessing in Python,” GeeksforGeeks, Oct. 29, 2017. https://www.geeksforgeeks.org/data-analysis/data-preprocessing-machine-learning-python/

[5] Zach, “K-Means Clustering in Python: Step-by-Step Example,” Statology, Aug. 31, 2022. https://www.statology.org/k-means-clustering-in-python/

[6] J. Chen, “Seaborn Swarmplot Axis Interval Formatting,” Stack Overflow, Jul. 31, 2020. https://stackoverflow.com/questions/63196379/seaborn-swarmplot-axis-interval-formatting